Intrusion Detection System using Soft Computing

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1. Introduction

With the rapid growth of internet-based systems and cloud infrastructure, cyber attacks have become increasingly frequent and sophisticated. Detecting malicious network activities in real time is a major challenge due to the large volume and complexity of network traffic data.

Intrusion Detection Systems (IDS) are designed to monitor network traffic and identify suspicious activities. Traditional machine learning techniques often struggle to capture complex non-linear intrusion patterns, resulting in missed attacks and false alarms.

This project focuses on developing an Intrusion Detection System using Machine Learning and Soft Computing techniques. Logistic Regression and Artificial Neural Network (ANN) models are implemented and compared using the NSL-KDD dataset.

The objective is to improve attack detection capability, reduce false negatives, and analyze the effectiveness of soft computing approaches in intrusion detection.

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2. Dataset Description

The project uses the NSL-KDD dataset, a widely used benchmark dataset for intrusion detection research.

The dataset contains:

- Network traffic records
- 41 features per record
- A class label indicating normal or attack traffic

The features include:

- Basic connection features
- Content-based features
- Traffic-based features

Attack categories include:

- Denial of Service (DoS)
- Probe
- Remote to Local (R2L)
- User to Root (U2R)

For this project, the problem is converted into binary classification:

- Normal Traffic → 0
- Attack Traffic → 1

The dataset is highly imbalanced, making false negative reduction an important challenge.

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3. Methodology

The system follows a structured machine learning pipeline consisting of preprocessing, model training, evaluation, and deployment.

3.1 Data Preprocessing

Several preprocessing techniques are applied to prepare the dataset for machine learning.

Steps Performed:

- Removed unnecessary columns
- Converted categorical features using One-Hot Encoding
- Applied feature scaling using StandardScaler
- Performed Train-Test Split (80:20)

Categorical features encoded:

- protocol_type
- service
- flag

The preprocessing pipeline ensures compatibility with machine learning algorithms and improves model performance.

3.2 Logistic Regression

Logistic Regression is implemented as the baseline traditional machine learning model.

Key Characteristics:

- Linear classification algorithm
- Suitable for binary classification
- Fast and computationally efficient

Threshold tuning and class balancing techniques are applied to improve Recall and reduce False Negatives.

Final Performance:

- Accuracy: 97.43%
- Recall: 97.53%
- AUC Score: 0.9959

The model achieved strong attack detection capability but produced relatively higher false positives.

3.3 Artificial Neural Network (ANN)

An Artificial Neural Network (ANN) is implemented using MLPClassifier as the soft computing approach.

ANN Architecture:

- Input Layer
- Hidden Layer (100 neurons)
- Output Layer

Advantages:

- Learns complex non-linear intrusion patterns
- Captures hidden feature relationships
- Improves attack classification capability

Final Performance:

- Accuracy: 99.67%
- Recall: 99.58%
- AUC Score: 0.9994

ANN significantly reduced false negatives and improved overall intrusion detection performance.

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4. Results and Performance Evaluation

The models are evaluated using:

- Accuracy
- Recall
- Confusion Matrix
- ROC Curve
- AUC Score

4.1 Logistic Regression Results

Confusion Matrix:

- False Positives: 355
- False Negatives: 291

The model achieved high recall after threshold tuning but still produced moderate false alarms.

4.2 ANN Results

Confusion Matrix:

- False Positives: 34
- False Negatives: 49

ANN drastically reduced missed attacks and improved detection capability.

4.3 Model Comparison

Metric	Logistic Regression	ANN
Accuracy	97.43%	99.67%
Recall	97.53%	99.58%

False Negatives	291	49
False Positives	355	34
AUC Score	“0.9959”	“0.9994”

The results clearly show that ANN outperformed Logistic Regression in all major evaluation metrics.

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5. Visualization and Dashboard

Several visualizations are used to analyze model performance:

- Confusion Matrix
- ROC Curve
- Recall Comparison
- False Negative Comparison
- AUC Analysis

An interactive dashboard is developed using Streamlit.

Dashboard Features:

- User input interface
- Real-time intrusion prediction
- Logistic Regression prediction
- ANN prediction
- Model selection support

The dashboard demonstrates practical deployment of the intrusion detection system.

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6. Conclusion

This project successfully demonstrates the application of Machine Learning and Soft Computing techniques for intrusion detection.

Logistic Regression provided strong baseline performance with high recall after optimization. However, the Artificial Neural Network significantly improved overall classification performance by learning complex non-linear intrusion patterns.

ANN achieved:

- Higher Recall
- Lower False Negatives
- Lower False Positives
- Better AUC Score

The results confirm that soft computing approaches are more effective for intrusion detection compared to traditional linear models.

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7. Future Work

Future improvements may include:

- Deep Learning based IDS models
- Real-time packet capture and monitoring
- Cloud deployment
- Spark Streaming integration
- Explainable AI for attack interpretation
- Full feature deployment using all 41 features

These enhancements can further improve scalability, efficiency, and real-world applicability of the system.
